DATABASE MANAGEMENT SYSTEMS

Dr. R. Vishnu Priya

Outline

- Introduction
  - File based approach
  - Database approach
  - Nuts and Bolts
- Database systems concepts
  - Data model
  - Three Tier architecture
  - Three schema architecture
  - Database schema – state – instance
  - DBMS languages
  - Over all Database Architecture
  - Advantage and Disadvantage

SAMPLE FILE – FRONT END

DreamHome
Property for Rent Details
Property Number: PG21

ddress 18 Dale Rd

ity Glasgow

ostcode G12

ype House **Rent** 600

o of Rooms 5

Allocated to Branch:
163 Main St, Glasgow

Branch No B003

Staff Responsible
Ann Beech

Owner's Details

ame Carol Farrel

ddress 6 Achray St,
Glasgow G32 9DX

el No. 0141-357-7419

Business Name

Address

Tel No.

DreamHome
Client Details
Client Number: CR74

First Name Mike **Last Name** Ritchie

Address 18 Tain St,
PA1G 1YQ

Tel No. 01475-392178

Property Requirement Details

Preferred Property Type House **Maximum Monthly Rent** 750

General Comments Currently living at home with parents
Getting married in August

Seen By Ann Beech **Date** 24-Mar-04

Branch No B003 **Branch City** Glasgow

SAMPLE FILE – BACK END

File based approach: example

File-based (pseudo-code)

find person (*name*:input, *record*:output)
begin
open *people*
repeat while *people* **has next**
  people → go to next record
  record := *people* → current record
  if *record* → *person* **is** *name*
      done
  else
      continue
  end
end
record := *invalid*
end

Database (SQL)

SELECT *
FROM people
WHERE
name = $NAME

Sales Department

DreamHome
Property for Rent Details
Property Number: PG21

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Branch No B003 **Branch City** Glasgow

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93

PrivateOwner

ownerNo	fName	lName	address	telNo
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025

Client

clientNo	fName	lName	address	telNo	prefType	maxRent
CR76	John	Kay	56 High St, London SW1 4EH	0207-774-5632	Flat	425
CR56	Aline	Stewart	64 Fern Dr, Glasgow G42 0BL	0141-848-1825	Flat	350
CR74	Mike	Ritchie	18 Tain St, PA1G 1YQ	01475-392178	House	750
CR62	Mary	Tregear	5 Tarbot Rd, Aberdeen AB9 3ST	01224-196720	Flat	600

Contracts Department

DreamHome
Lease Details
Lease Number: 10012

Client No. CR74

Full Name Mike Ritchie

Address (previous) 18 Tain St,
PA1G 1YQ

Tel No. 01475-392178

Property No. PG21

Address 18 Dale Rd,
Glasgow G12

Payment Details

Monthly Rent 600

Payment Method Cheque

Deposit 1200 **Paid (Y or N)** Y

Rent Start Date 1-Jul-04

Rent Finish Date 30-Jun-05

Duration 1 Year

Lease

leaseNo	propertyNo	clientNo	rent	payment Method	deposit	paid	rentStart	rentFinish	duration
10024	PA14	CR62	650	Visa	1300	Y	1-Jun-05	31-May-05	12
10075	PL94	CR76	400	Cash	800	N	1-Aug-05	31-Jan-05	6
10012	PG21	CR74	600	Cheque	1200	Y	1-Jul-05	30-Jun-05	12

PropertyForRent

propertyNo	street	city	postcode	rent
PA14	16 Holhead	Aberdeen	AB7 5SU	650
PL94	6 Argyll St	London	NW2	400
PG21	18 Dale Rd	Glasgow	G12	600

Client

clientNo	fName	lName	address	telNo
CR76	John	Kay	56 High St, London SW1 4EH	0171-774-5632
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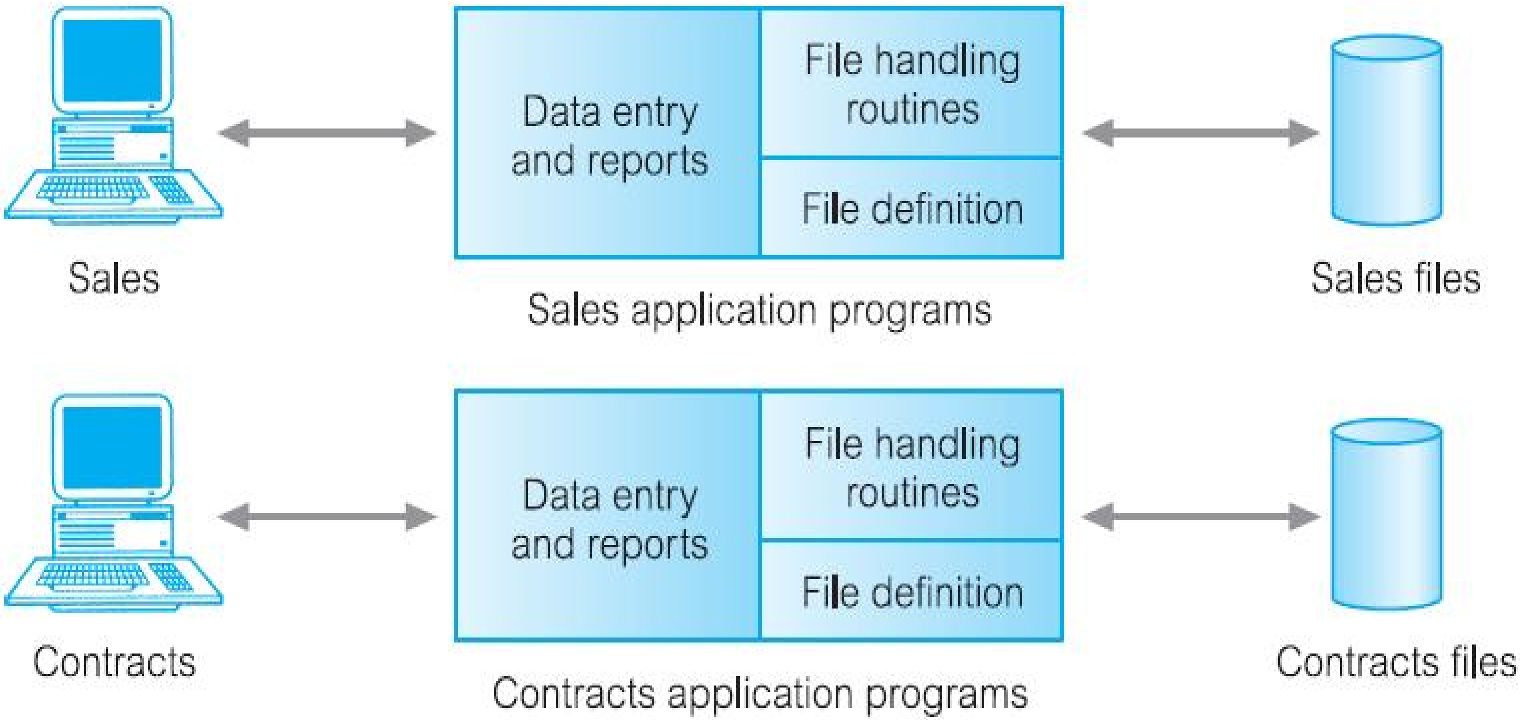
CR62	Mary	Tregear	5 Tarbot Rd, Aberdeen AB9 3ST	01224-196720



Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)

PrivateOwner (ownerNo, fName, lName, address, telNo)

Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)

PropertyForRent (propertyNo, street, city, postcode, rent)

Client (clientNo, fName, lName, address, telNo)

Limitation of File System

Separation and isolation of data

Duplication of data

Data dependence

Incompatible file formats

Fixed queries/proliferation of application programs

Isolation of data

When **data is isolated in separate files**, it is more difficult to access data.

- A list of all houses that match the requirements of clients

propertyNo	street	city	postcode	type	rooms	rent	ownerNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40
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Duplication of data

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Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)

PropertyForRent (propertyNo, street, city, postcode, rent)

Client (clientNo, fName, lName, address, telNo)

Update Problem

- Payroll Department

StaffSalary(staffNo, fName, lName, sex, salary, branchNo)

The Personnel Department also stores staff details, namely:

Staff(staffNo, fName, lName, position, sex, dateOfBirth, salary, branchNo)

Duplication Data Issues

- Duplication is wasteful. It **costs time and money** to enter the data more than once.
- It takes up additional **storage space**, again with associated costs. Often, the duplication of data can be avoided by sharing data files.
- Perhaps more importantly, duplication can lead to **loss of data integrity**; in other words, the data is no longer consistent. For example, consider the duplication of data between

Data dependence

- PropertyForRent

  Address field (40 to 41 characters)

- **open** the original **PropertyForRent** file for reading;
- **open a temporary** file with the new structure;
- **read** a record from the **original file**, **convert the data to** conform to the **new structure**, and
- **write** it to the **temporary** file. Repeat this step for all records in the original file;
- **delete** the **original** PropertyForRent file;
- **rename** the **temporary** file as PropertyForRent

Incompatible file formats

- Contracts Department wants to find the names and addresses of all owners whose property is currently rented out

Sales Files

PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)

PrivateOwner (ownerNo, fName, lName, address, telNo)

Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Contracts Files

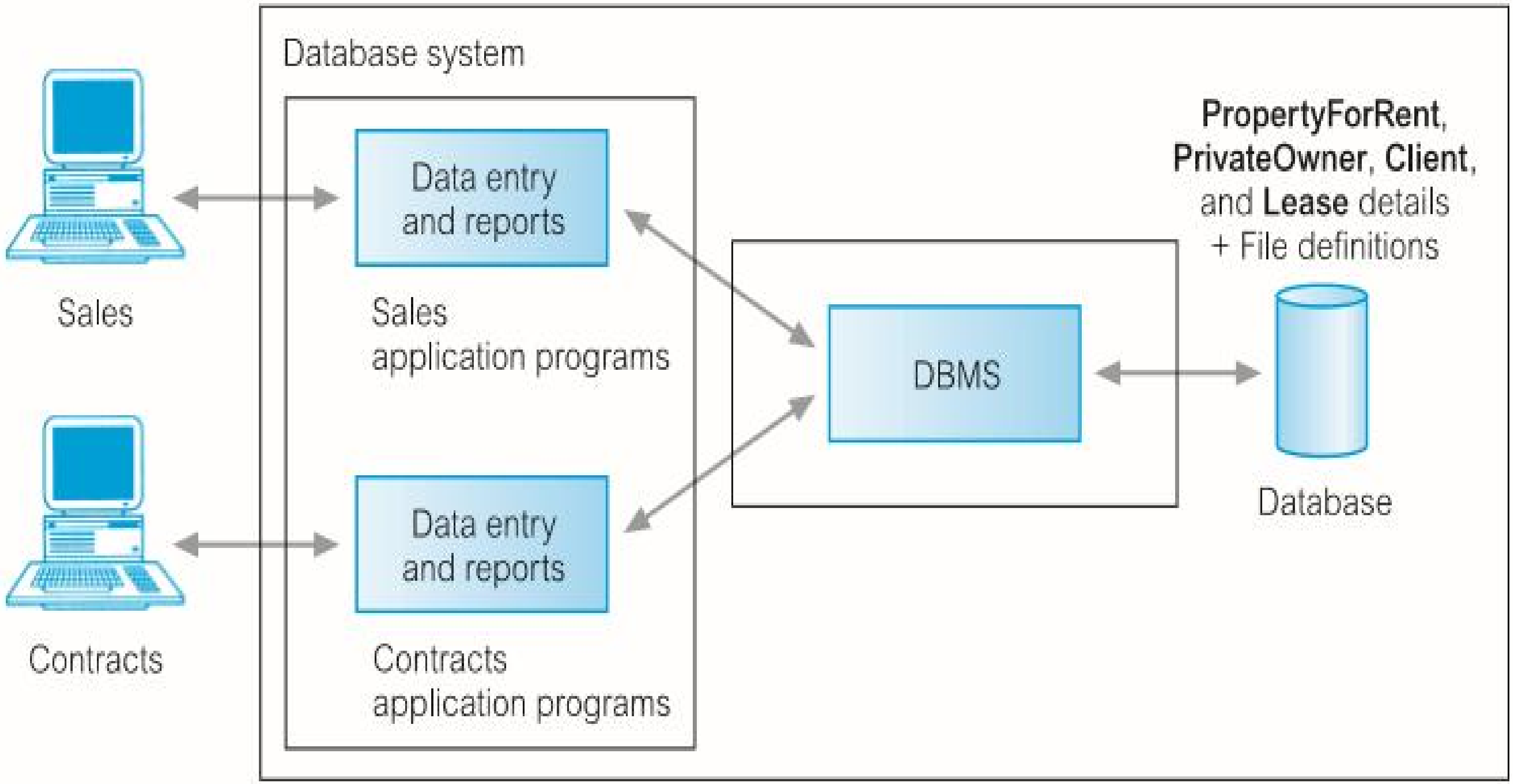
Lease (leaseNo, propertyNo, clientNo, rent, paymentMethod, deposit, paid, rentStart, rentFinish, duration)

PropertyForRent (propertyNo, street, city, postcode, rent)

Client (clientNo, fName, lName, address, telNo)

Fixed queries/proliferation of application programs

- There was no provision for **security or integrity;**
- **Recovery**, in the event of a hardware or software failure, was limited or non-existent;
- Access to the files was **restricted to one user** at a time – there was no provision for shared
- Access by staff in the same department.

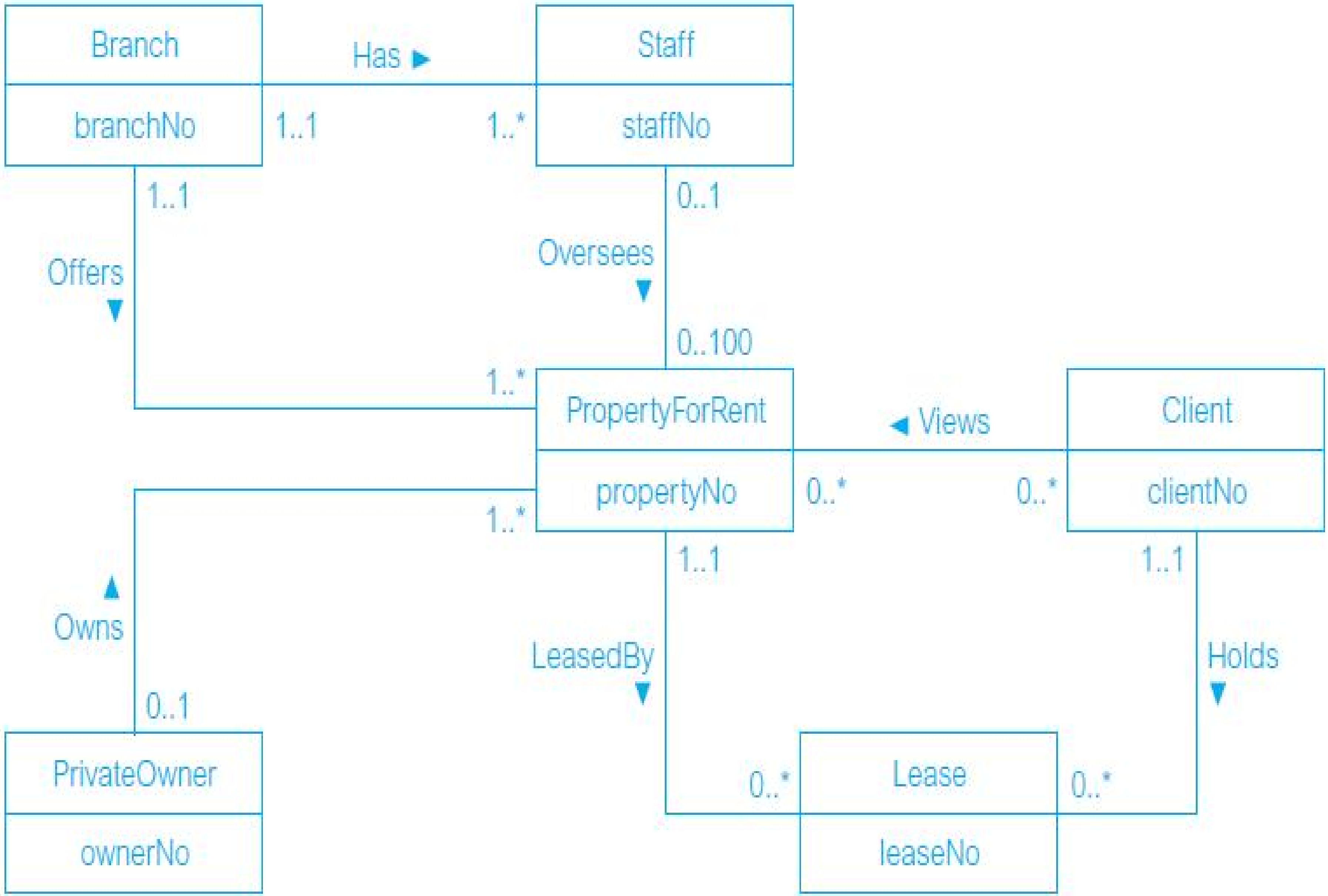


PropertyForRent (propertyNo, street, city, postcode, type, rooms, rent, ownerNo)

PrivateOwner (ownerNo, fName, lName, address, telNo)

Client (clientNo, fName, lName, address, telNo, prefType, maxRent)

Lease (leaseNo, propertyNo, clientNo, paymentMethod, deposit, paid, rentStart, rentFinish)


Branch
branchNo
Has ▶
1..1
1..*
Staff
staffNo
1..1
Offers
0..1
Oversees
0..100
1..*
PropertyForRent
propertyNo
◀ Views
0..*
0..*
Client
clientNo
1..*
Owns
0..1
PrivateOwner
ownerNo
1..1
LeasedBy
1..1
Holds
0..*
Lease
leaseNo
0..*

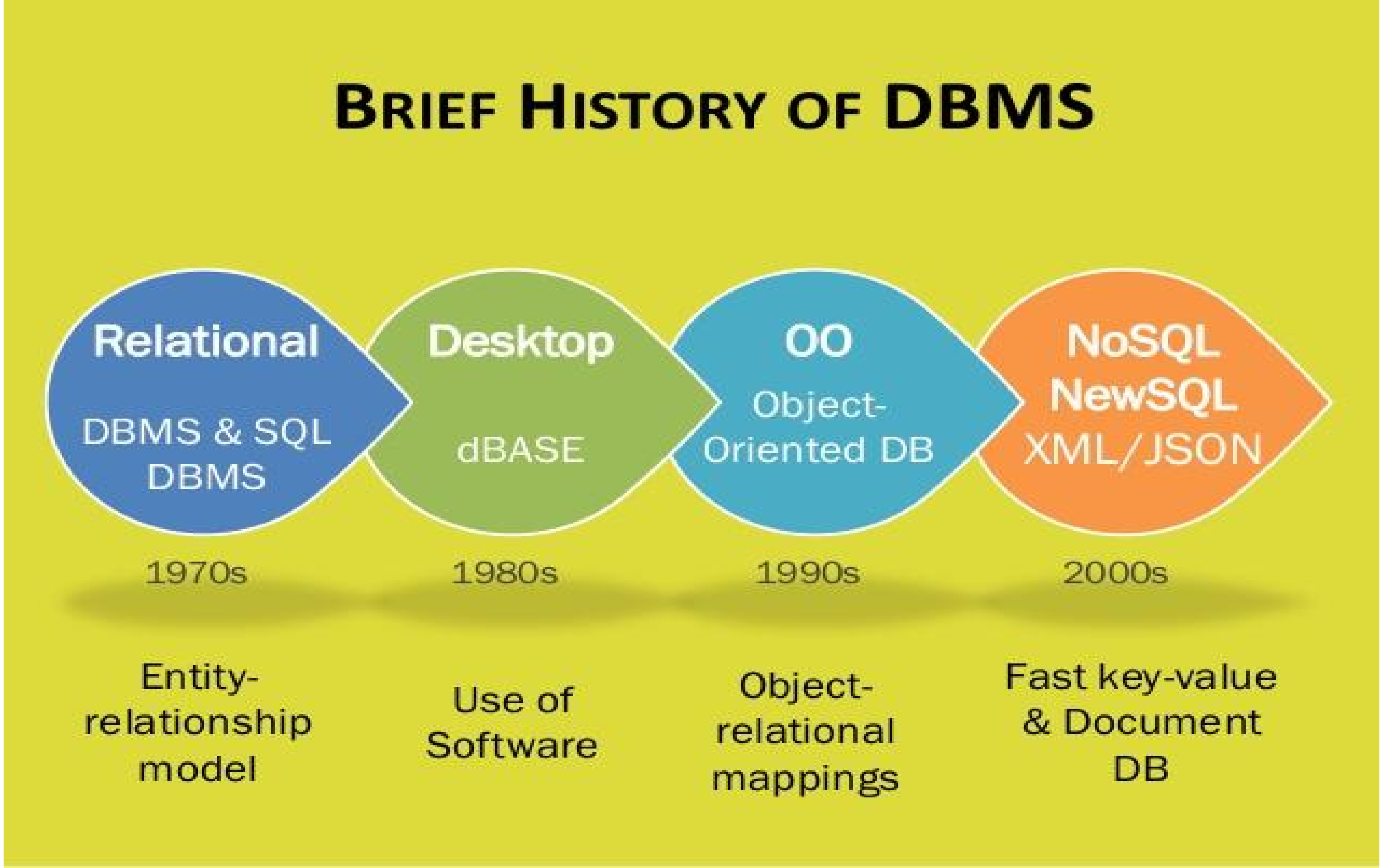
FILE Based Approach

- In case of college database, there may be the **number of applications** like **General Office, Library, Department Office, Hostel** etc. Each of these applications may maintain the following information into **own private file applications** in case of file management system:

General office	Libray	Hostel	Account office
Rollno	Rollno	Rollno	Rollno
Name	Name	Name	Name
Class	Class	Class	Class
Father_Name	Address	Father_Name	Address
Date of birth	Date_of _birth	Date_of_birth	Phone_No
Address	Phone_No	Address	Fee
Phone_No	No_of_books_issued	Phone_No	Installments
Previous_Record	Fine	Mess_Bill	Discount
Attendance	etc.	Room No	Balance
Marks		etc.	Total
etc.			etc.

It is clear from the above file systems, that there are some common data items of the student which has to be mentioned in each application, like **Rollno, Name, Class, Father_Name, Address, Phone_No, Date_of_birth etc.** which are stored repeatedly in file system in each application. It will cause the problem of redundancy which results in wastage of storage space and difficulty in maintenance.

History of DBMS



History (cont.)

Pre-relational DB (70)

- First distinction between logical and physical data management
- Data management becomes important.
- Need of data models

Relational Database Systems (80)

- No redundancy in data storage
- Integrated data control even by distributed data storage
- Multiuser operation and high performance

Post-relational Databases (90)

- objectoriented Databases
- Multidimentional Databases
- Datawarehouses
- Distributed databases

Introduction to Database Approach

Database: A collection of related data.

Data: Known facts that can be recorded and have an implicit meaning.

Mini-world: Some part of the real world about which data is stored in a database. For example, student grades and transcripts at a university.

Database Management System (DBMS): A software package/ system to facilitate the creation and maintenance of a computerized database.

Database System: The DBMS software together with the data itself. Sometimes, the applications are also included.

Data Models

- Data Model: A set of concept to describe the **structure of database and certain constraints** the database should obey.

- A collection of tools for describing
  - Data
  - Data relationships
  - Data semantics
  - Data constraints
- **Relational model**
- **Entity-Relationship data model** (mainly for database design)
- **Object-based data models** (Object-oriented and Object-relational)
- **Semi structured data model** (XML)
- **Other older models**:
  - Network model
  - Hierarchical model

Schemas, instances and database state

In any data model it is important to distinguish between the *description* of the data and the *data itself*:

- ***Database schema***: description of a database, which is specified during database design and is not expected to change too frequently.
  = the information stored in the catalog
- ***Database state***: the data in the database at a particular moment, also called the current set of *instances*.
  = the data currently in the database

Sample extract of a university database (1)

Database schema

STUDENT (number, name, address, email)

COURSE (number, name)

BUILDING (number, address)

...

Database state

STUDENT

Number	Name	Address	Email
0165854	John Doe	154 West Str.	John.doe@student.kuleuven.be
0168975	Fred Astere	7 Scene Field Str.	Fred.astere@student.kuleuven.be
0157895	Kyara Welton	89 Fifth Av.	Kyara.welton@student.kuleuven.be

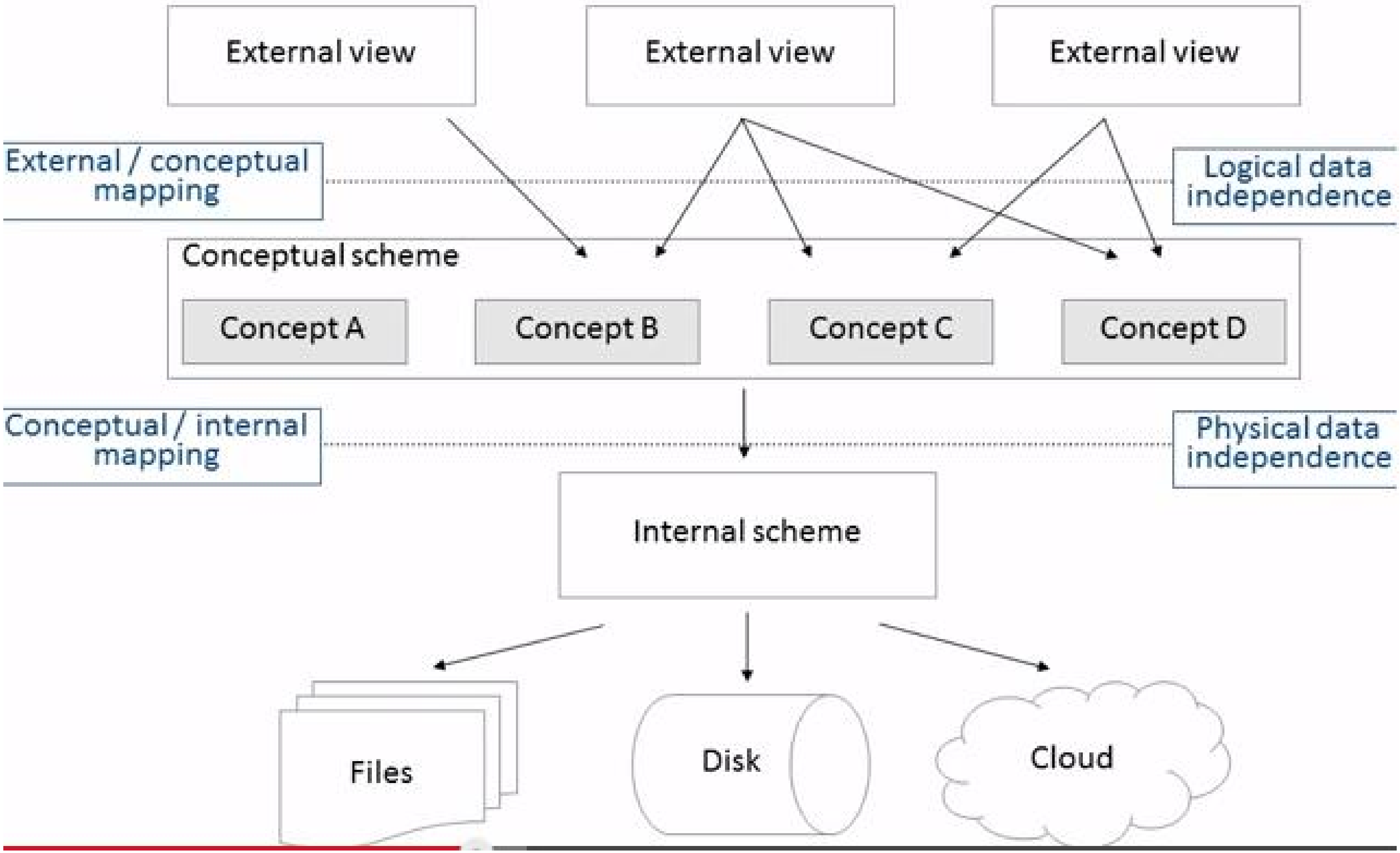
COURSE

Number	Name
D0I69A	ICT Service Management
D0R04A	Strategic management
D0T21A	Macro economics

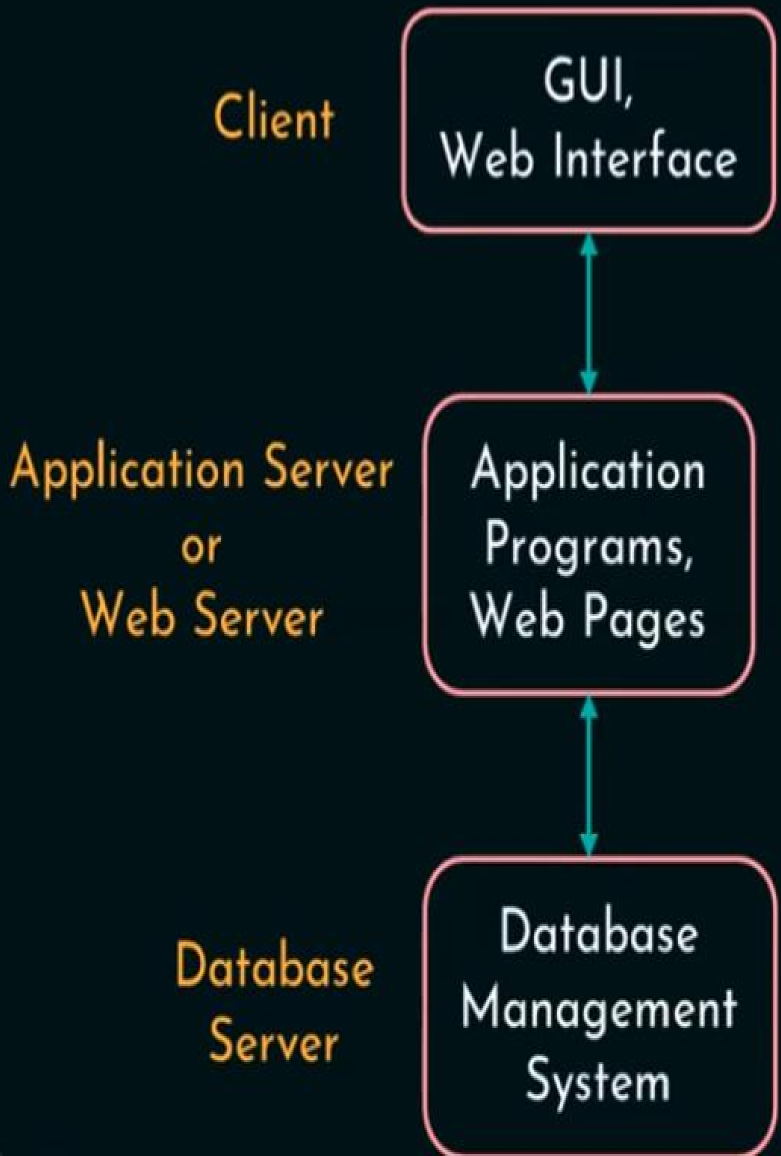
BUILDING

Number	Address
0589	Naamsestraat 69
0365	Naamsestraat 78
0589	Tiensestraat 115

The three-schema architecture



Three Tier Architecture



- ☆ Predominant software architecture.
- ☆ 3 logical and physical tiers.
- ☆ Each tier runs on its own infra.
- ☆ Can be developed simultaneously.
- ☆ Can be scaled and updated.
- ☆ Presentation, Application and Data tiers.
- ☆ Advantages:
  - Faster development,
  - Improved scalability, reliability and security.

Three Tier Architecture

Client / Front end

- ☆ Top level tier.
- ☆ User Interface (UI).
- ☆ Application level.
- ☆ Collect and display info.
- ☆ Desktop applications or GUI.
- ☆ HTML, CSS, Java Script etc.

Three Tier Architecture

Application or Web Server

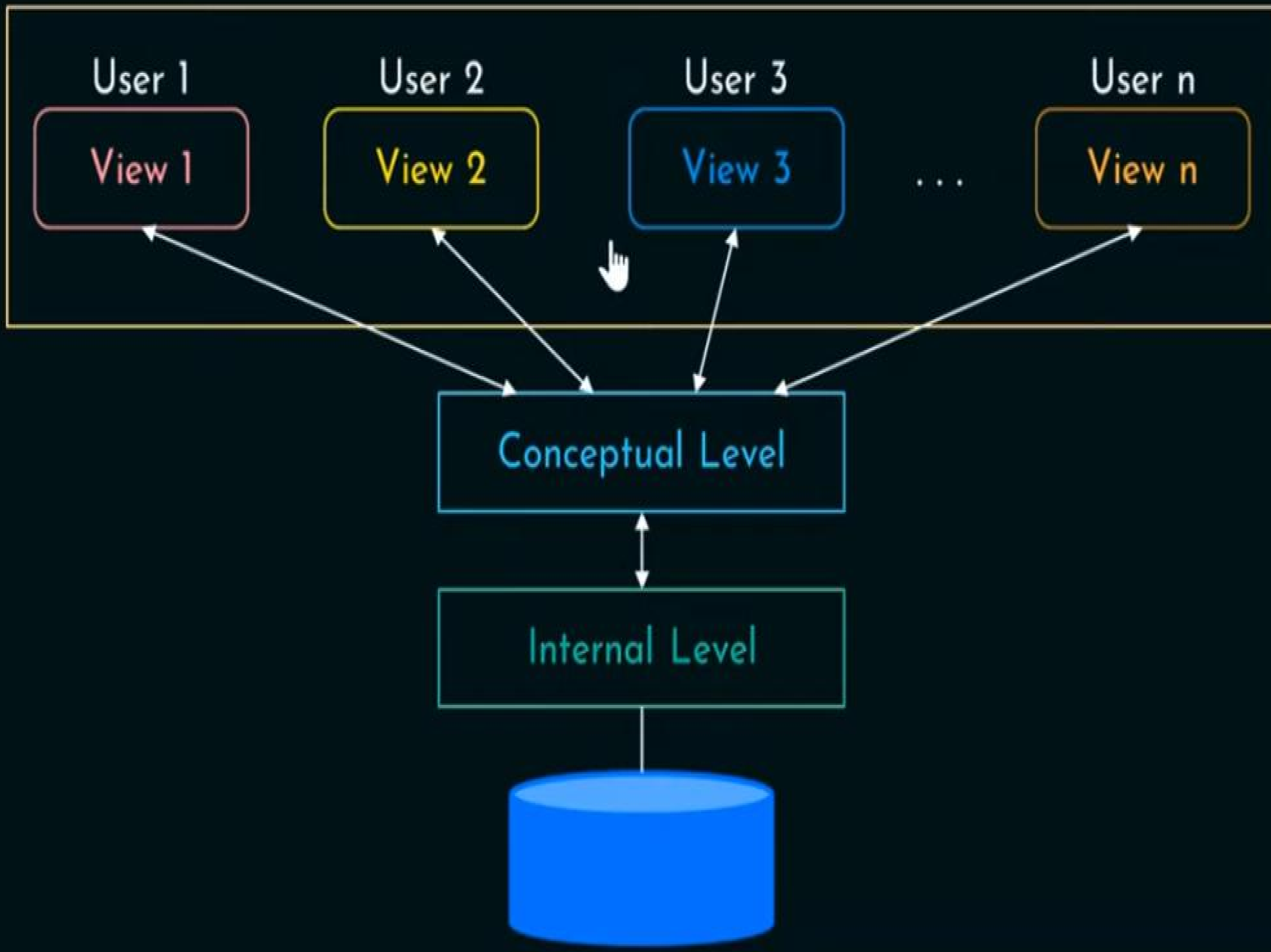
- ☆ Logic or middle level tier.
- ☆ Heart of the application.
- ☆ Information is processed.
- ☆ Business logic and rules.
- ☆ Add, delete or update data in the data tier.
- ☆ Python, Java, Perl, PHP, Ruby etc.,
- ☆ Communicates with data tier using API calls.

Three Tier Architecture

Backend / Database server

- ☆ Database tier or backend.
- ☆ Data are stored and managed.
- ☆ Communication - presentation and data tier.
- ☆ RDBMS - MySQL, MariaDB, Oracle, DB2, Microsoft SQL server, PostgreSQL etc.
- ☆ DBMS - Cassandra, CouchDB, MongoDB etc.

Generalized DBMS Three-tier Architecture

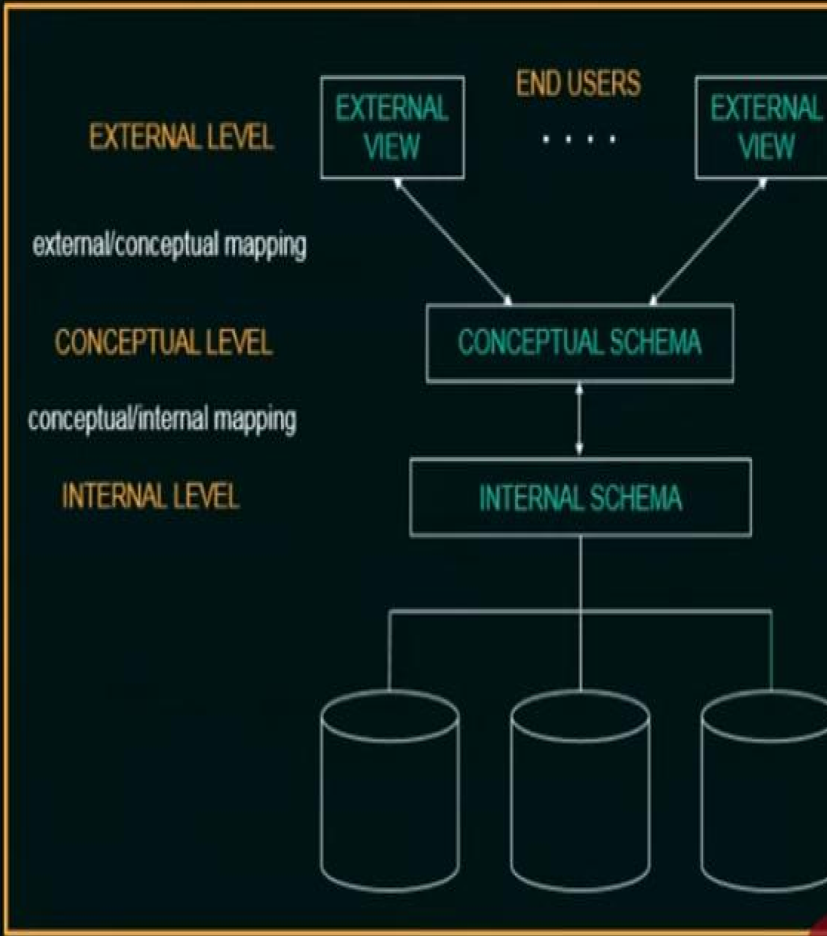


❖ Three-Schema Architecture

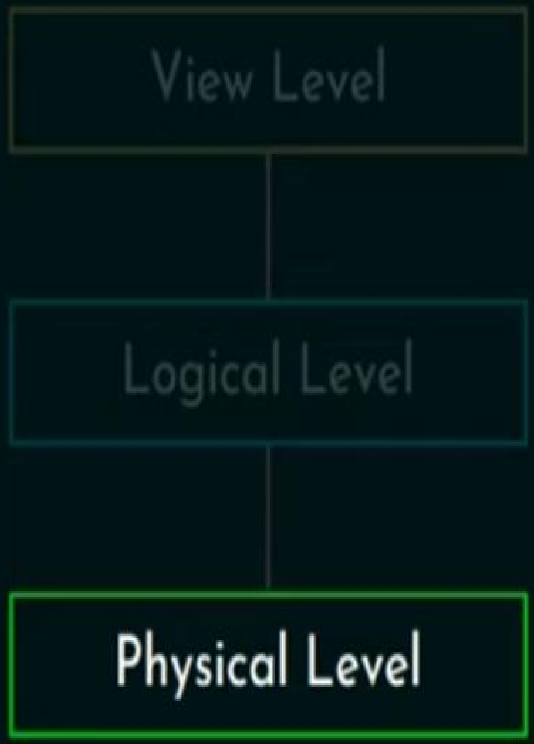
★ **Goal:** to separate the user applications and the physical database.

★ **3 Levels:**

- **Internal Level:**
  - ➔ Describes the physical storage structure of the database.
  - ➔ Describes complete details of data storage and access



View of Data



Physical Level

- ☆ Lowest level of abstraction.
- ☆ Deals with how.
- ☆ Complex low level data structures.
- ☆ Database level.
- ☆ Storage level.

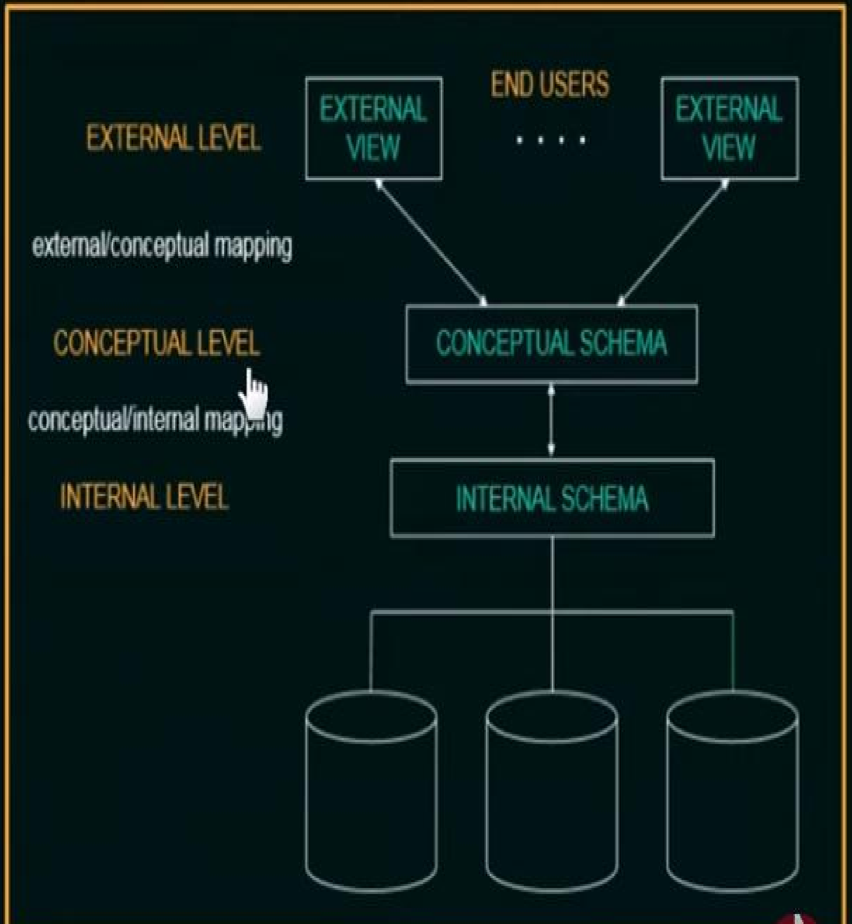
Physical Schema

☆ Example:

```
type Student = record
        Rollno  : numeric (5);
        Name    : char (25);
        Class   : char (10);
end;
```

❖ Three-Schema Architecture

- Conceptual Level:
  - ➔ Hides the details of the physical storage structure and concentrates on describing entities, data types, relationships, constraints, etc.



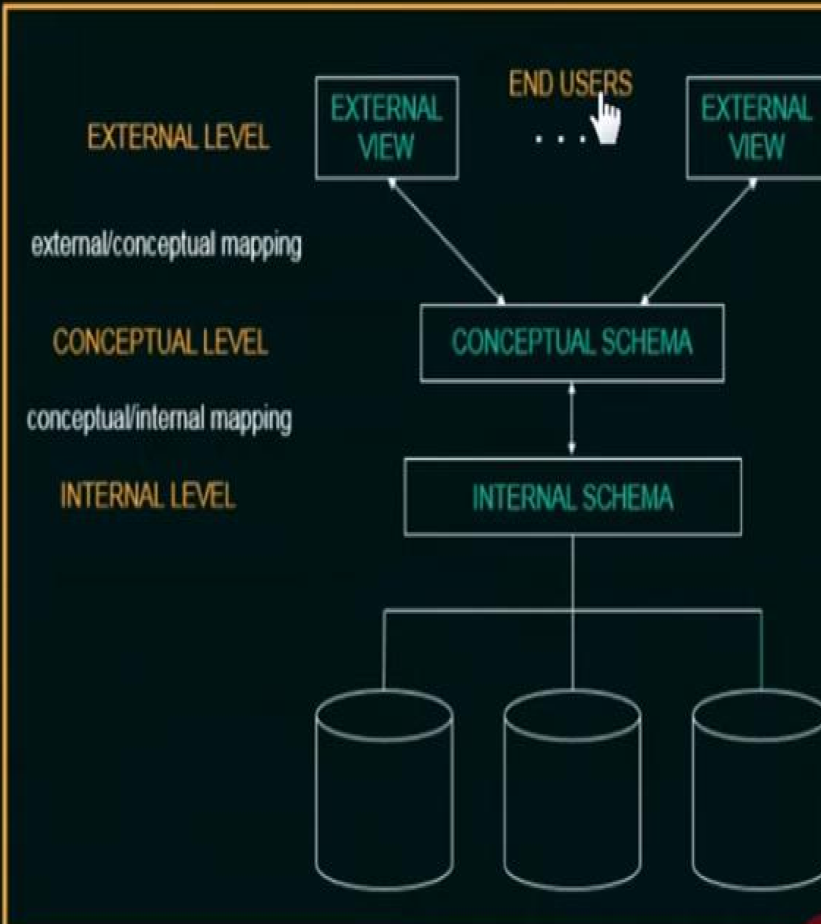
View of Data

Logical Level

- ☆ Deals with what and relationship.
- ☆ Entire database with simple data structures.
- ☆ Physical Data Independence.

❖ Three-Schema Architecture

- External Level:
  - ➔ Describes the part of the database that a user is interested in and hides the rest of the database from the user group.



View of Data

View Level

Logical Level

Physical Level

View Level

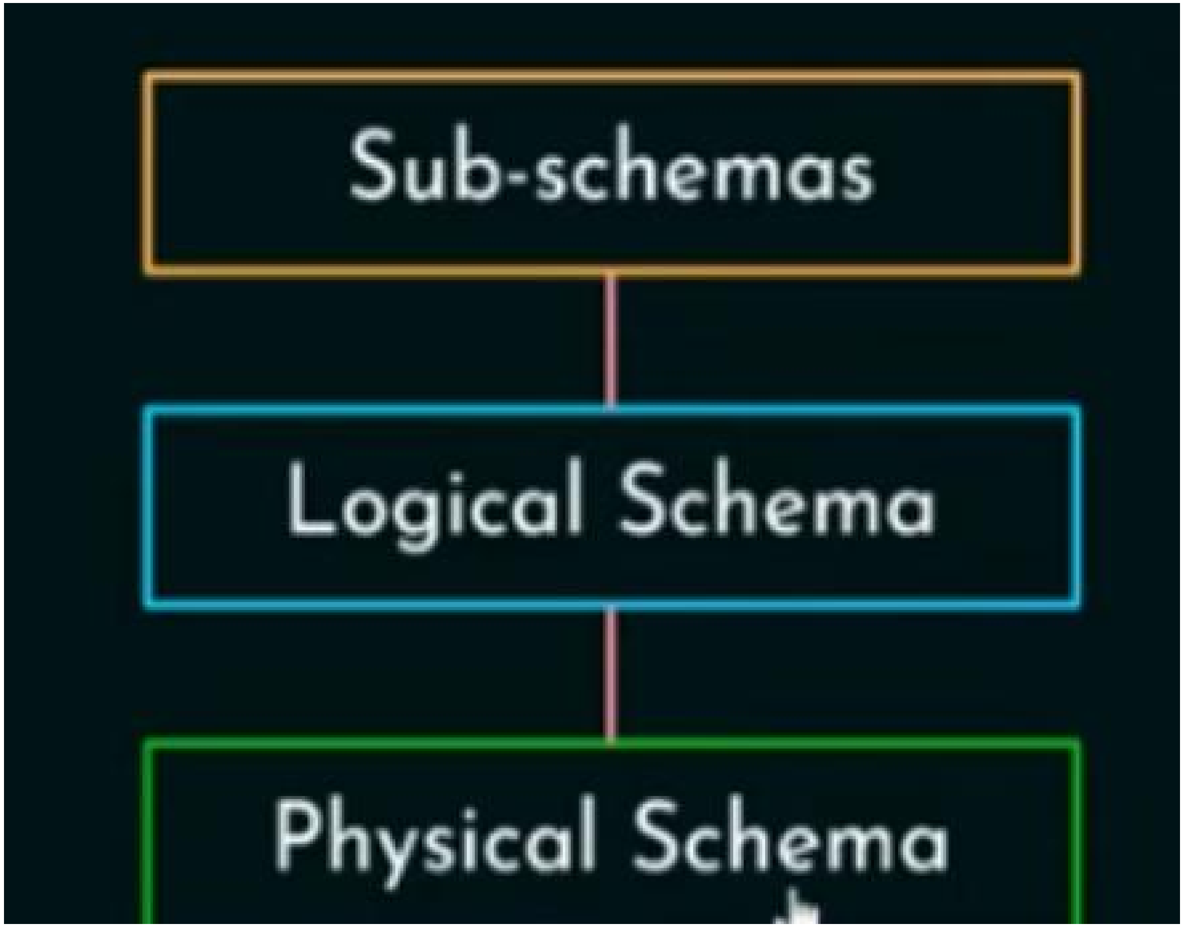
☆ Highest level of abstraction.

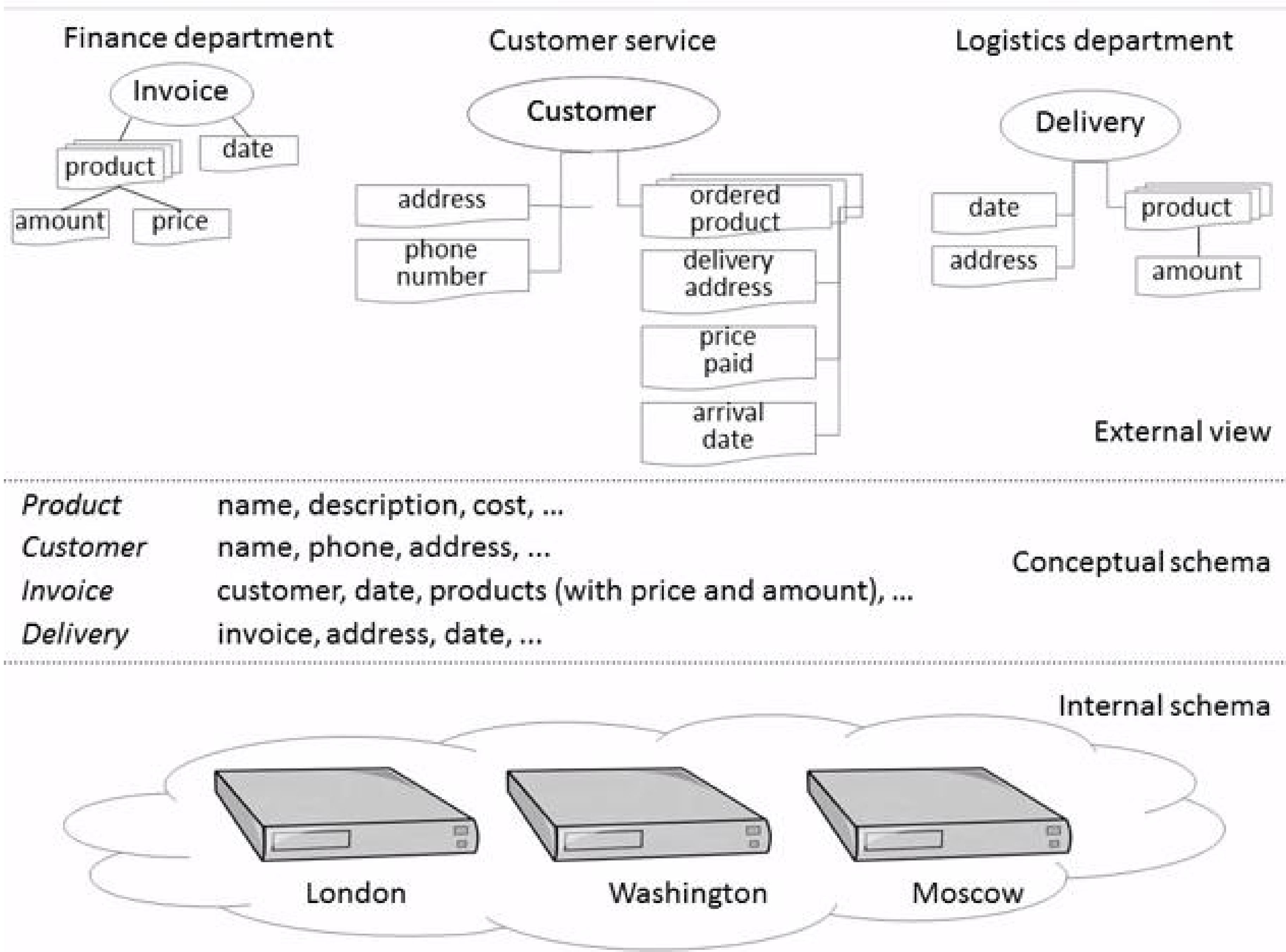
☆ Users and access.

☆ Interaction with the system.

☆ Application programs.

☆ Several views and security.


Sub-schemas
Logical Schema
Physical Schema


Finance department
Invoice
product
date
amount
price
Customer service
Customer
address
phone number
ordered product
delivery address
price paid
arrival date
Logistics department
Delivery
date
address
product
amount
External view
Product
name, description, cost, ...
Customer
name, phone, address, ...
Invoice
customer, date, products (with price and amount), ...
Delivery
invoice, address, date, ...
Conceptual schema
Internal schema
London
Washington
Moscow

Data Independence

Physical data independence

It means we change the **physical** storage/level without affecting the conceptual or external view of the **data**. The new changes are absorbed by mapping techniques.

Logical data independence is the ability to modify the **logical** schema without causing application program to be rewritten.

DBMS languages

- *DBMS Languages*
  - ***Data Definition Language (DDL):*** language used by the database administrator (DBA) to define the database's conceptual, internal and external schemas
  - ***Data Manipulation Language (DML):*** language used to retrieve, insert, delete and modify data. DML statements can be entered interactively from a terminal or embedded in a general-purpose programming language.

Commands	Statements	Description
Data Definition Language	Create , Alter Drop, Rename	Sets up ,changes and removes data structures called tables
Data Manipulation Language	Insert Delete Update Select	Add, removes and changes rows in Db. Retrieve data from Db.
Data Control Language	Grant ,revoke	Give or removes data access to others
Transaction Control Language	Commit, Rollback and save point	Manage the changes made by DML

Database Users

1. Naive users.
2. Application programmers.
3. Sophisticated users.
4. Specialized users.

1. Naive Users

- ☆ Unsophisticated users.
- ☆ Invoke/use the application program.
- ☆ Ex: Clerk, Teller etc.
- ☆ Web/Mobile/UI interface.
- ☆ Desktop applications.

2. Application Programmers

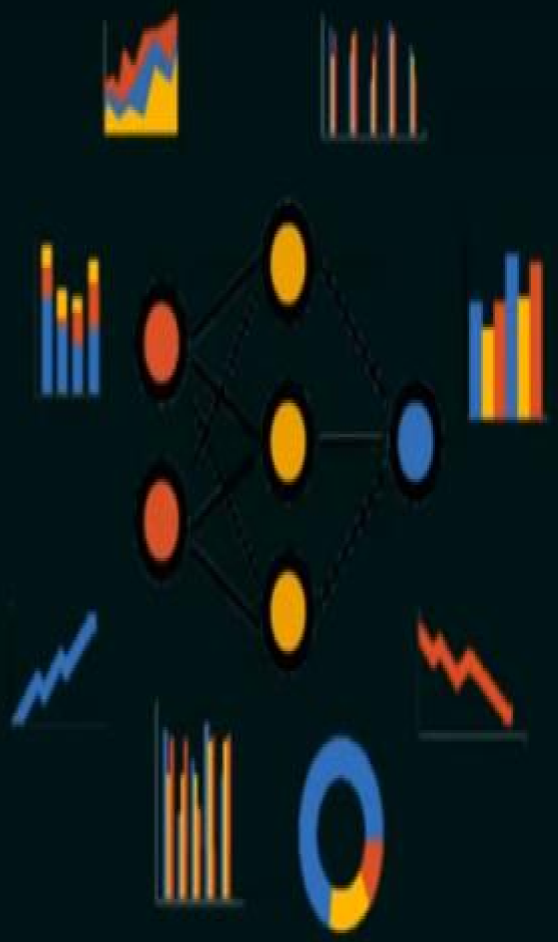
- ☆ Computer professionals.
- ☆ Application development tools.
- ☆ Develop user interfaces.
- ☆ RAD.
- ☆ Web/Mobile interface.

3. Sophisticated Users

- ☆ Interact with system.
- ☆ Database query languages.
- ☆ Data analysis software.
- ☆ Data/Business analysts.

4. Specialized Users

- ☆ Specialized DB applications.
- ☆ Computer aided design systems.
- ☆ Knowledge base and expert Systems.
- ☆ Multimedia data.
- ☆ Next generation.



Database Administrator

- ☆ DBA.
- ☆ Data and programs access those data.
- ☆ Central control.
- ☆ Key role with complete privilege.

Functions of DBA

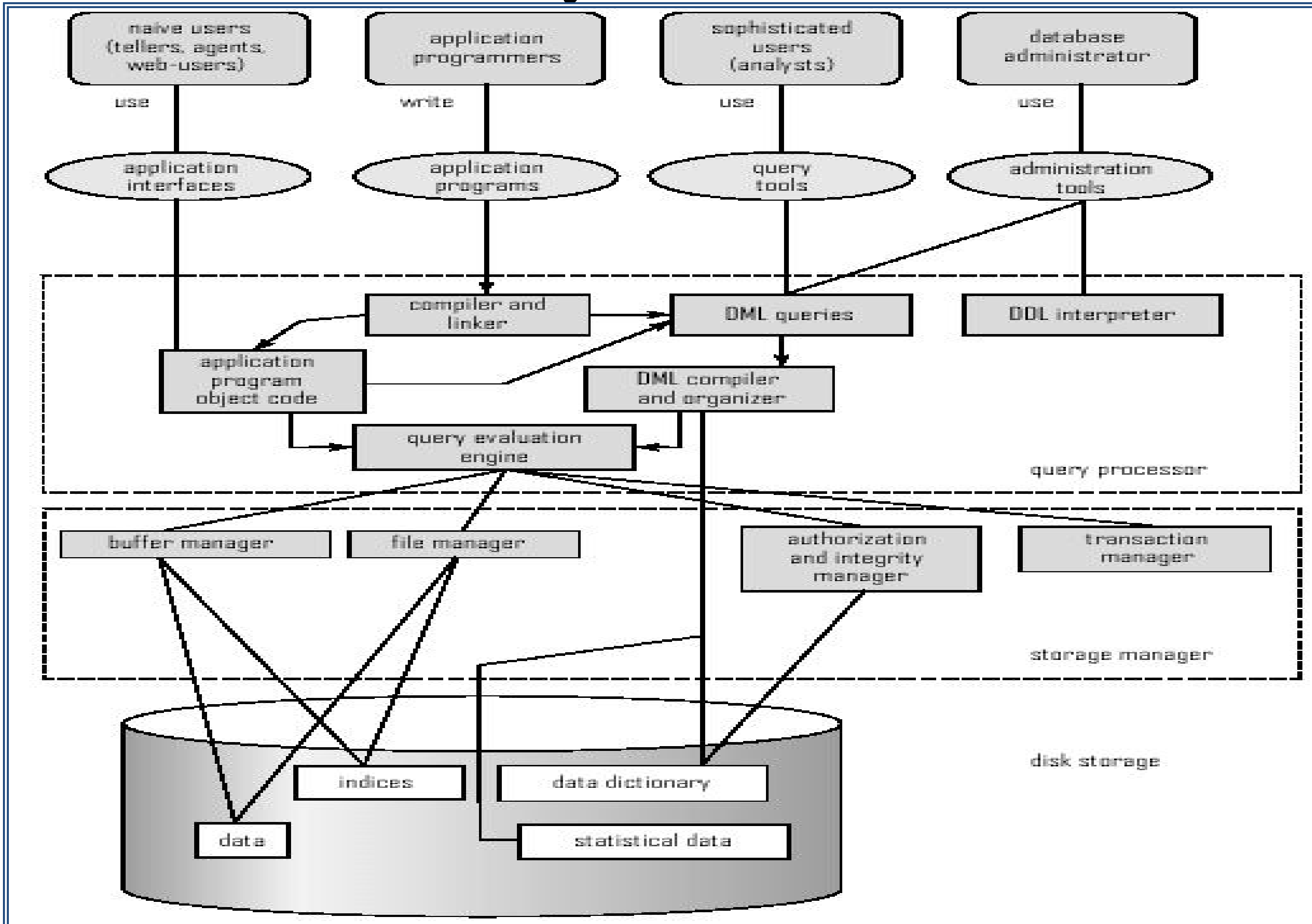
- ☆ Schema definition.
- ☆ Storage structure and access method definition.
- ☆ Schema and physical-organization modification.
- ☆ Granting of authorization for data access.
- ☆ Routine maintenance.
  - Periodic backup.
  - Disk space management.
  - Performance.

Database Architecture

The architecture of a database systems is greatly influenced by the underlying computer system on which the database is running:

- Centralized
- Client-server
- Parallel (multiple processors and disks)
- Distributed

Overall System Structure

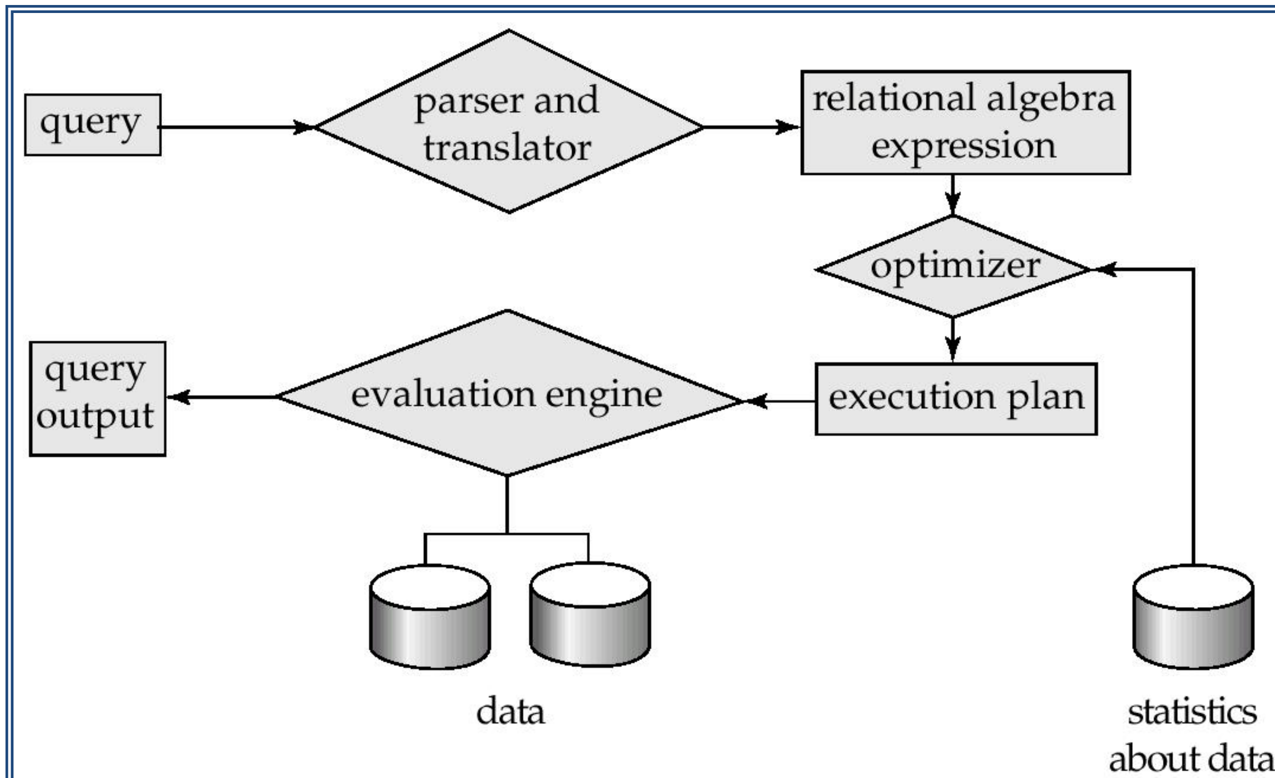


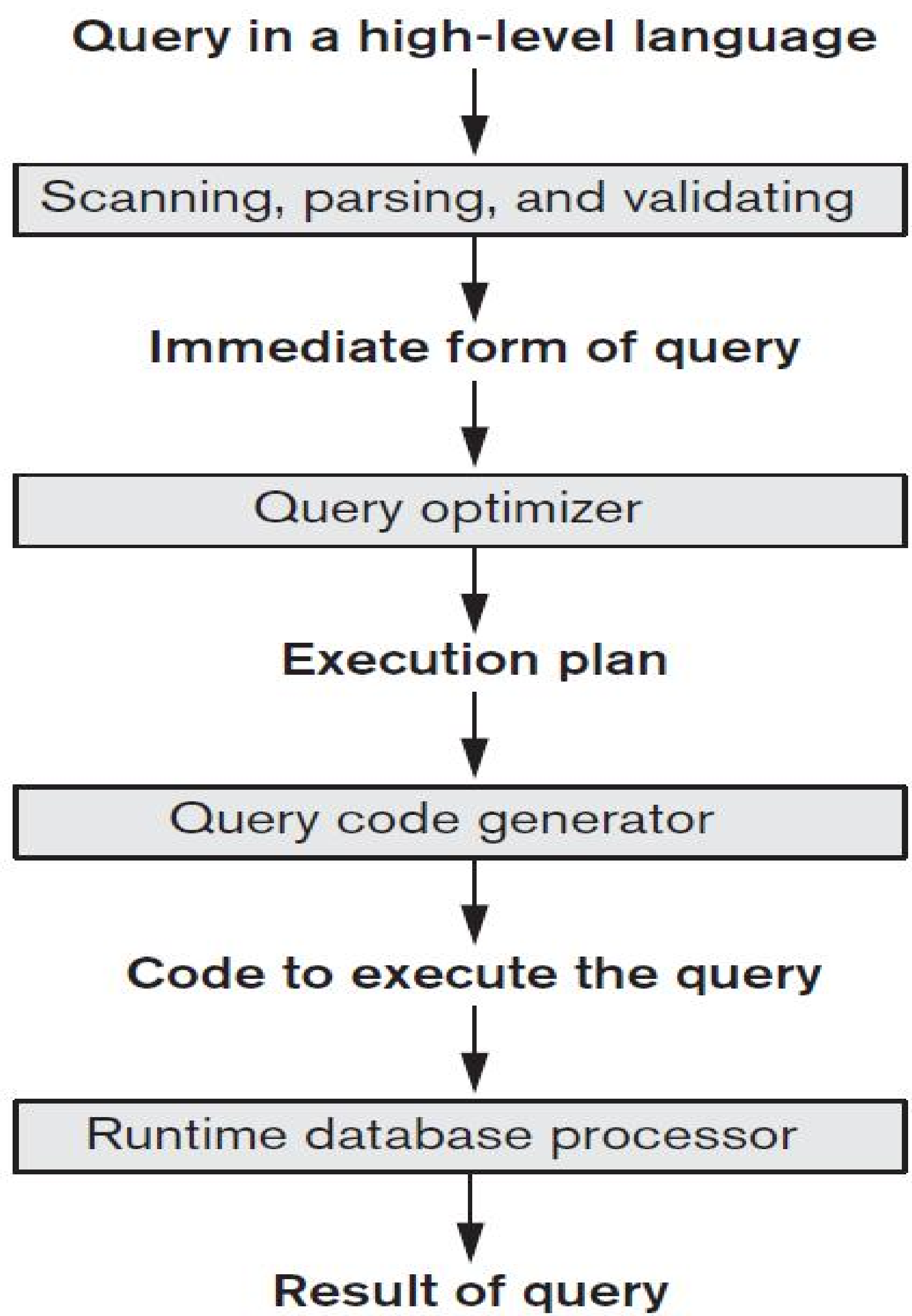
Data storage and Querying

- Query processing
- Transaction processing
- Storage management

Query Processing

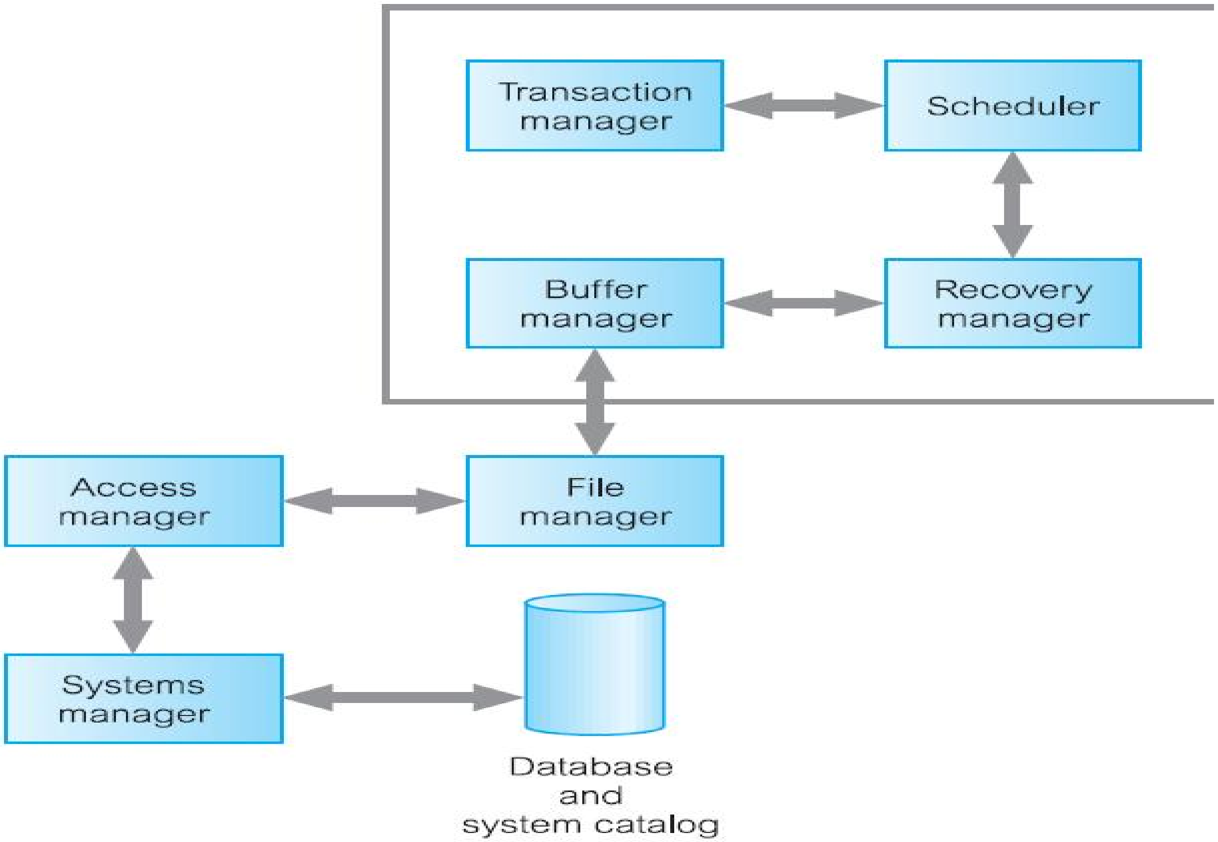
1.Parsing and translation
2. Optimization
3. Evaluation




Query in a high-level language
Scanning, parsing, and validating
Immediate form of query
Query optimizer
Execution plan
Query code generator
Code to execute the query
Runtime database processor
Result of query

- S**canner** identifies the query tokens—such as SQL keywords, attribute names, and relation names—that appear in the text of the query
- **Parser** checks the query syntax to determine whether it is formulated according to the syntax rules (rules of grammar) of the query language.
- **Validated** by checking that all attribute and relation names are valid and semantically meaningful names in the schema of the particular database being queried.
- An internal representation of the query is then created, usually as a tree data structure called a **query tree**. It is also possible to represent the query using a graph data structure called a **query graph**.
- The DBMS must then devise an **execution strategy** or **query plan** for retrieving the results of the query from the database files.
- A query typically has many possible execution strategies, and the process of choosing a suitable one for processing a query is known as

DBMS transaction subsystem



- The **transaction manager** coordinates transactions on behalf of application programs. It communicates with the **scheduler**, the module responsible for implementing a particular strategy for concurrency control.
- The scheduler is sometimes referred to as the **lock manager** if the concurrency control protocol is locking based. The objective of the scheduler is to maximize concurrency without allowing concurrently executing transactions to interfere with one another, and so compromise the integrity or consistency of the database.
- If a failure occurs during the transaction, then the database could be inconsistent. It is the task of the **recovery manager** to ensure that the database is restored to the state it was in before the start of the transaction, and therefore a consistent state.

- Finally, the **buffer manager** is responsible for the efficient transfer of data between disk storage and main memory.

Advantage and Disadvantage

1. **Controlling Data Redundancy:**

File: Duplicated copies of the same data are created at many places

DBMS: Data of an organization is integrated into a single database

Data Consistency:

By controlling the data redundancy, the data consistency is obtained. If a **data item appears only once, any update** to its value has to be performed only **once** and the updated value (new value of item) is immediately available to all users.

Data Sharing:

Data can be **shared by authorized users** of the organization. The DBA manages the data and gives rights to users to access the data. Many users can be authorized to access the same set of information simultaneously. The remote users can also share same data. Similarly, the data of same database can be shared between different application programs.

Data Integration:

In DBMS, data in database is stored in tables. A single database contains multiple tables and **relationships can be created between tables** (or associated data entities). This makes easy to retrieve and update data.

5. Integrity Constraints:

Integrity constraints or consistency **rules can be applied to database** so that the correct data can be entered into database. The constraints may be applied to data item within a single record or they may be applied to relationships between records.

Examples:

The ***examples of integrity*** constraints are:

- (i) 'Issue Date' in a library system cannot be later than the corresponding 'Return Date' of a book.
- (ii) Maximum obtained marks in a subject cannot exceed 100.
- (iii) Registration number of BCS and MCS students must start with 'BCS' and 'MCS' respectively etc.

Standard constraints that are intrinsic in most of the DBMS

Constraint Name	**Description**
PRIMARY KEY	Designates a column or combination of columns as Primary Key and therefore, values of columns cannot be repeated or left blank.
FOREIGN KEY	Relates one table with another table.
UNIQUE	Specifies that values of a column or combination of columns cannot be repeated.
NOT NULL	Specifies that a column cannot contain empty values.
CHECK	Specifies a condition which each row of a table must satisfy.

For **example,** when you draw amount from the bank through **ATM card**, then your account balance is compared with the amount you are drawing. If the amount in your **account balance is less** than the amount you want to **withdraw**, then a **message is displayed** on the screen to inform you about your account balance.

6. Data Security:

Data security is the protection of the database from unauthorized users. Only the authorized persons are allowed to access the database. Some of the users may be allowed to access only a part of database i.e., the data that is related to them or related to their department. Mostly, the DBA or head of a department can access all the data in the database. Some users may be permitted only to retrieve data, whereas others are allowed to retrieve as well as to update data. The database access is controlled by the DBA. He creates the accounts of users and gives rights to access the database. Typically, users or group of users are given usernames protected by **passwords.**

For example, if you have an account of e-mail in the "**hotmail.com**" (a popular website), then you have to give your correct username and password to access your account of e-mail. Similarly, when you insert your ATM card into the **Auto Teller Machine (ATM)** in a bank, the machine reads your ID number printed on the card and then asks you to enter your pin code (or password). In this way, you can access your account.

7. Data Atomicity:

A **transaction** in commercial databases is referred to as **atomic unit** of work. For example, when you **purchase** something from a point of sale (POS) terminal, a number of tasks are performed such as;

- **Company stock is updated.**
- **Amount is added in company's account.**
- **Sales person's commission increases etc**.

All these tasks collectively are called an atomic unit of work or transaction. These **tasks must be completed in all**; otherwise partially completed tasks are **rolled back**. Thus through DBMS, it is ensured that only consistent data exists within the database.

8. Database Access Language

Most of the DBMSs provide **SQL as standard database access language**. It is used to access data from multiple tables of a database.

9. Development of Application:

The **cost and time** for developing **new applications** is also **reduced**. The DBMS provides tools that can be used to develop application programs. For example, some wizards are available to generate Forms and Reports. Stored procedures (stored on server side) also reduce the size of application programs.

10. Creating Forms:

Form is very important object of DBMS. You can create Forms very easily and quickly in DBMS, Once a Form is created, it can be used many times and it can be modified very easily. The created Forms are also saved along with database and behave like a software component.

A **Form provides very easy way** (user-friendly interface) to enter data into database, edit data, and display data from database. The non-technical users can also perform various operations on databases through Forms without going into the technical details of a database.

11. Report Writers:

Most of the DBMSs provide the **report writer tools** used to create reports. The users can create reports very easily and quickly. Once a report is created, it can be used many times and it can be modified very easily. The created reports are also saved along with database and behave like a software component.

12. Control Over Concurrency:

In a computer file-based system, if two users are allowed to access data simultaneously, it is possible that they will interfere with each other. For example, if both users attempt to perform update operation on the same record, then one may overwrite the values recorded by the other. Most DBMSs have sub-systems to control the concurrency so that transactions are always recorded" with accuracy.

13. Backup and Recovery Procedures:

In a computer file-based system, the user creates the backup of data regularly to protect the valuable data from damaging due to failures to the computer system or application program. It is a time consuming method, if volume of data is large. Most of the DBMSs provide the 'backup and recovery' sub-systems that automatically create the backup of data and restore data if required. For example, if the computer system fails in the middle (or end) of an update operation of the program, the recovery sub-system is responsible for making sure that the database is restored to the state it was in before the program started executing.

14. Data Independence:

The separation of data structure of database from the application program that is used to access data from database is called data independence. In DBMS, **database and application programs are separated from each other**. The **DBMS sits in between** them. You can easily change the structure of database without modifying the application program. For example you can modify the size or data type of a data items (fields of a database table).

On the other hand, in computer file-based system, the structure of data items are built into the individual application programs. Thus the data is dependent on the data file and vice versa.

Disadvantages of Database Management System

1. Cost of Hardware & Software:
2. Cost of Data Conversion
3. Cost of Staff Training
4. Appointing Technical Staff
5. Database Failures